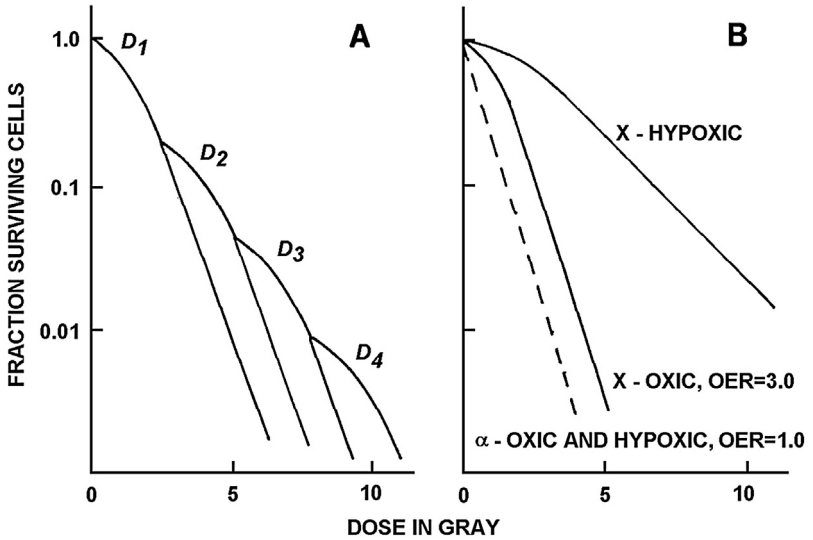


Figure 15.5 Fraction of surviving mammalian cells as a function of dose. A) Effect of dose fractionation. B) Irradiation by X-rays and by α-particles under aerated and hypoxic conditions.

This result suggests that the addition of a reductant would reduce the radiation effect which is also observed. For example, 20 Gy is required to kill 80% of a cell population in the presence of cysteine (or cysteamine), while in the reference system without cysteine the same effect is caused by only 5 Gy. Cysteine is oxidized to cystine.

Several other *radiation protection agents* are known. These compounds are typically amino thiols, similar to the natural amino acid cysteamine. They probably function as scavengers for the products of water radiolysis. Their effectiveness is evaluated by determination of the *dose reduction factor* (DRF), which is the ratio of $LD_{50/30}$ for protected and unprotected animals. Because of their chemical toxicity, many can only be administered in small doses.

No evidence has been found for cells or of higher organisms supporting the development of long-lasting *radiation resistance*. Certain bacteria have been shown to develop a seeming resistance to radiation after receiving small radiation doses over a long period of time. However, this is possibly due to the formation of mutated organisms with a different sensitivity to radiation than the original ones.

15.6. EPIDEMIOLOGICAL OBSERVATIONS OF EFFECTS AT LARGE RADIATION DOSES

We may distinguish between accidental (e.g. Japan and Chernobyl) and deliberate (as given in radiotherapy) exposures to large radiation doses. The former are said to be

stochastic because the harm caused is statistically distributed over the exposed population; the frequency of tumor induction is assumed to increase linearly with the dose. Deliberate large irradiations are *deterministic* because the damage is caused intentionally to a certain organ or population; such irradiations are considered to have a *threshold value*, below which no effects are observed.

15.6.1. Radiation sickness after accidental exposure

Very large instantaneous doses (>10 Gy) occur in explosions of nuclear weapons, in accidents involving nuclear reactors or criticality accidents, or from carelessness in working with accelerators, X-ray equipment or radioactive installations (e.g. ^{60}Co sources used for technical and therapeutic purposes), criticality accidents, and in handling unshielded strong radiation sources or unshielded radioactive waste. Such doses are very unlikely to be received in work involving amounts of ≤ 1 GBq of radioactivity.

Instantaneous whole body doses (i.e. those received within a few hours) of $\gg 10$ Sv lead to death within 24 h through destruction of the neurological system. At $7-8$ Sv, the main damage is to the gastrointestinal tract, causing severe bleeding which may lead to death within several days to a month. Doses <0.5 Sv are rarely lethal. For doses between these two levels, intensive hospitalization is required for survival. At the higher end of this range, death usually occurs from 4 to 8 weeks after the accident through infection as because of the destruction of the leukocyte ("white blood cells") forming organs. Those surviving this period usually recover completely. For doses <0.5 Sv, the only proven effect is a decrease in the white blood count (leukopenia). The threshold value for early somatic damage for short irradiation times appears to be about 0.25 Sv.

An example of the effects of high doses was the Tokai-mura criticality accident in Japan in 1999 where 3 workers in a nuclear facility got 4, 8 and 18 Sv respectively. The person receiving the highest dose died within 3 weeks due to a completely destroyed immune system. The man receiving 8 Sv died after 7 months due to important organ failure as a result of the irradiation. The man receiving 4 Sv went back to normal life after 3 months.

The most common type of overexposure in radioactive work involves high instantaneous doses to the hands. Fortunately, the hands, of which the skin is the most sensitive tissue, can stand fairly large doses (Table 15.7). If they do receive extremely high doses ($\gg 10$ Sv β or γ), amputation is usually required. Although in some cases, skin transplants have provided temporary relief.

15.6.2. Radiation therapy and deterministic studies

Radiation therapy usually consists of the delivery of large instantaneous or fractionated doses to tissues for which surgical operation is impossible or undesirable. The effects of large organ doses are well described in the literature and is one of the main forms of

Table 15.7 Observed effects of instantaneous radiation doses to the hands

<2 Gy	No proven effect
~4 Gy	Erythema, skin scaling, follicle deaths
6–7 Gy	Skin reddening after a few hours, which then decreases, later strongly increases after 12–14 d, then finally disappears within a month; pigmentation
>8.5 Gy	As above, but irreversible degeneration of the skin is visible to the naked eye (the skin becomes hard and cracked); degeneration of the binding tissue with increasing dose
50–80 Gy	Development of non-healing skin cancer; amputation necessary

tumor treatment. Radioactive ^{60}Co (up to 200 TBq), ^{137}Cs (up to 20 TBq) γ-sources, or braking radiation from small linear electron accelerators are used for treatment of deeply located organs; otherwise, X-rays are used more commonly. The organ doses for malignant tumors are usually <100 Sv for mature breast, prostate and blood; typically $\leq$50 Sv for other internal organs and skin (often given in a series of smaller doses) and 10–20 Sv for the gonads, breast and bone marrow. In comparison, diagnostic radiation doses are $\leq$ 0.1 Sv. The use of internal radiation sources for therapy is described in §18.5.4.

A special form of radiation therapy uses heavy high energy ions from accelerators. The decreasing velocity of charged particles in matter results in a very high specific ionization near the end of the path (the Bragg peak, Fig. 7.7.b). The energy of the particles is selected according to the depth and type of tissue to be penetrated so that the particles have the proper range to provide a very high local dose in the proper volume of the sick tissue. This technique (the "proton knife") has been particularly important in treating diseases of the pituitary gland, which is located deep inside the brain.

In some cases, local irradiation of some tissues is produced by the use of radioactive nuclides implanted in the tissue by means of needles or small capsules. For example, needles of ^{90}Sr–^{90}Y, pellets of ^{198}Au, etc., have been implanted in the pituitary gland (for acromegalia, Cushing's disease, and cancer), in the breast (for breast cancer), in the prostate, and in the nerves (to reduce pain). The local dose may be 100's of Sv. Also, radioactively labeled tumor seeking compounds are used.

It should be noted that a patient's response to a certain radiation dose is highly individual; thus, all patients cannot be treated alike for the same tumor. Statistically up to 20% of the patients may be "lost" if this individuality is neglected! It is explained by the different "normal levels" of the patient's immune defense, which may have a hereditary cause. Small radiation doses seem to activate the immune response system by stimulating antigen production. This is used in radiation therapy by exposing the patient to a comparably small "pre-treatment" dose (~0.5 Gy) before the large organ doses are delivered in order to hasten the recovery of the patient after the treatment. The immune system then rapidly takes care of the radiation products.

15.6.3. Stochastic cancer induction

The first *radiation induced cancer* in a human was reported in 1902 when skin cancer was observed on the hand of a radiologist working with X-rays. During the following decades, quite a number of such cases were reported and occurred in either persons employed in medical radiology or in patients treated with radiation for benign lesions (e.g. eczema, birth marks or tuberculous lymph nodes). Other types of radiation induced cancers were reported as thyroid cancer and sarcoma in bone or soft tissues. The cumulative doses were usually quite high (several 10ths of Gy). Today this type of radiation induced cancer is rather uncommon due to increased knowledge in radio-physics and clinical radiobiology.

The A-bomb survivors in Japan are the most important source of information on human whole-body irradiation. The population was large and varied as regards age and gender, the time of exposure was short and well defined, and it has been possible to make reliable estimates of the dose to each individual. The most recent evaluation (BEIR V, the so-called DS86 dosimetry) comprises 76 000 persons, who had been exposed to $\leq 7-8$ Gy of instantaneous gamma and neutron irradiation. Their health and their children's health have now been followed in detail for 50 years. Figure 15.6 shows in (A) the chromosomal aberration frequency, and in (B) the number of leukemias observed as a function of dose. The excess leukemia frequency is 3.16 by the original 1965 estimation (T65) and 3.10 by the new 1986 dosimetry (DS86). The excess malignancies are rather low for other types of cancer.

The raw data in Figure 15.6.A show "best" lines for the Hiroshima and the Nagasaki victims; the central line is a common fit. The difference between Hiroshima and Nagasaki is believed to be due to a much higher contribution of neutrons in Hiroshima due to the construction of the bomb; Nagasaki was probably only gamma radiation effects. This is in agreement with the higher damage produced by high-LET radiation in Figures 15.4 and 15.5.B.

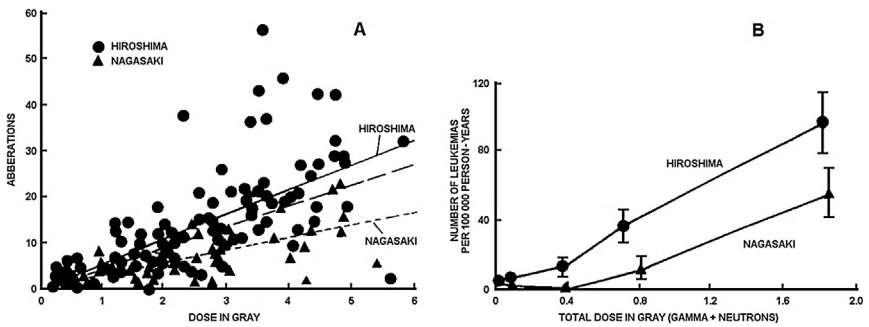


Figure 15.6 Radiation effects on population in Hiroshima and Nagasaki as a function of dose. A. Chromosome abberations. B. Leukemia frequency.

The total leukemia frequency is five times lower for the Japanese bomb victims who received only a slightly larger radiation dose than the average of the whole Japanese population. This may be caused by an early removal of "leukemia sensitive" individuals (i.e. pre-destined victims of leukemia because of inherited leukemia oncogenes) from the group or "protection by radiation" related to the stimulation of the DNA repair mechanism as discussed above, which reduces the susceptibility to the normal incidence of leukemia.

It is important to bear in mind that the number of radiation induced cancers in this cohort (a group of persons with a common statistical characteristic) is relatively small and thus the statistical uncertainty is large: in 1950–1985 a total of 5,936 cases of cancer were reported compared to the statistically expected 5,596 in an unirradiated reference group of similar sex-age composition, i.e. an excess of only 6% (350 cases).

Following large dose exposure, *leukemia* appears first after an approximately short latent period of about 2–3 years. The incidence of leukemia reaches peak frequency around 6–8 years and then declines and almost disappears about 25 years after exposure. This is shown for the population exposed after the Hiroshima and Nagasaki bombings, see Figure 15.7. Cancers others than leukemia, however, tend to exhibit a different behavior. They appear after a latent period of 10 years following the exposure and then show progressive increase with time. The *breast* seems to be one of the most susceptible organs for radiation induced cancer, but the A-bomb cohorts indicate a risk only in the

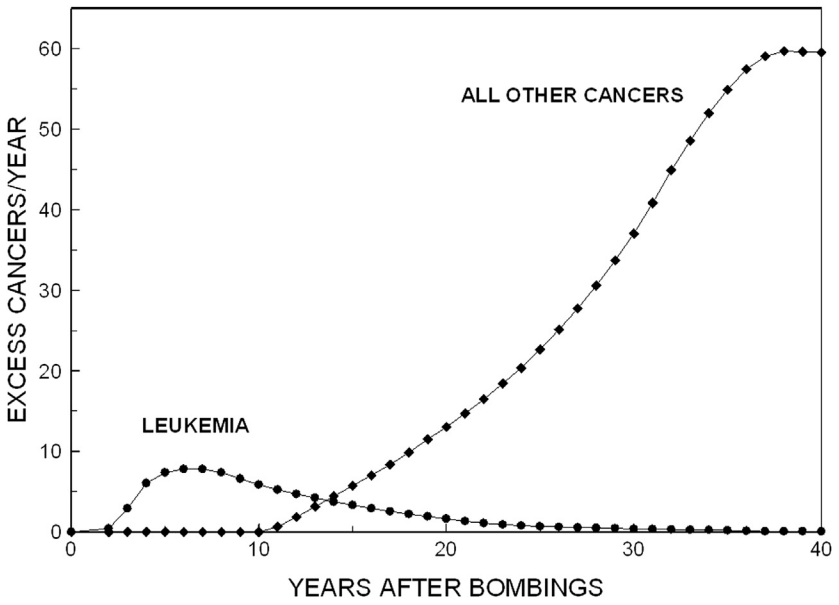


Figure 15.7 Excess cancer incidence rates in the japanese population irradiated by the A-bombs *(After Kato & Shimizi and the RERFoundation).*

younger age group (<40 y at exposure). Except for the breast, the *thyroid* is the most susceptible organ in humans for radiation induced solid tumors. This is the only cancer induction (an 80-fold increase among children) observed so far among the Chernobyl victims. The mortality risk, however, is very low because most of these tumors are benign. *Lung cancer* can be induced by γ-rays, but interest is presently focused on the risk from radon daughters in air. *Bone sarcoma* may develop after local exposure to X-rays or γ-rays. The latency period is 5−10 years for large doses (20−70 Gy).

For the Japanese survivors, increased detection of leukemia (13 times higher than naturally expected), multiple myeloma (6 times), cancer in the colon, urinary tract and breast (about twice) have been reported. In comparing an observed frequency of cancer with an expected one, it is important to correct for differences in sex and age groups.

Information on radiation effects from ingestion or inhalation of large amounts of radioactive substances is mainly limited to four cases: (i) uranium mine workers who inhaled and ingested Rn and Rn-daughters; (ii) the painters of luminous dials in Europe (before 1930, they ingested radium while sharpening their brushes by licking them); (iii) people living where the air has a high Rn concentration; (iv) patients treated by high amounts of ^{131}I for thyroid disorders (a non-stochastic investigation). The first two groups have been followed for more than five decades with the lifetime risks for fatal cancers estimated as 0.05% per Sv for bone sarcoma and 0.2% per Sv for cancer of the bone marrow. Radon is discussed in §§15.10, 15.11.1 and 15.13.5.

Saenger et al. (1968) studied 36 000 patients with hyperthyroidism (a cancer causing the thyroid to grow), of which 22 000 were treated with ^{131}I (only β, no γ; local doses of several hundred Sv) while the rest underwent surgery or chemical therapy. Although the ^{131}I patients received bone marrow doses of about 100 mSv, no difference was observed in relation to the non-irradiated group with respect to incidence of leukemia.

In 1988 Holm et al. reported a similar comprehensive study of 35 000 patients who had been administered 2×10^6 Bq ^{131}I each, which caused a thyroid dose of about 0.5 Sv per person. The mean time of examination was 44 years for patients of all ages. In this group 50 thyroid cancers were discovered (compared to the expected 40 cases for untreated), indicating a relative risk of 1.27 ± 0.33 (95% confidence interval). The excess of 10 thyroid cancers is so small that it falls within statistical uncertainty. Thus, Holm et al. conclude that "these data provide little proof that ^{131}I is carcinogenic in humans" at these doses and dose rates. The examples illustrate the difficulty in proving carcinogenesis even at relatively high doses.

Information on the effects of large "whole-body internal" radiation doses comes mainly from laboratory experiments on animals. Figure 15.8 shows the excess tumor frequency for mice irradiated with doses up to 120 Gy in a comparison of the effects of different radiation sources: A, ^{90}Sr-induced osteosarcomas in female CBA mice (Nilsson 1975); B, bone tumors in humans from incorporated ^{226}Ra (Rowland 1971); C, kidney tumors in rats by X-radiation (Maldague 1969); D, skin tumors in rats by electrons (Burns

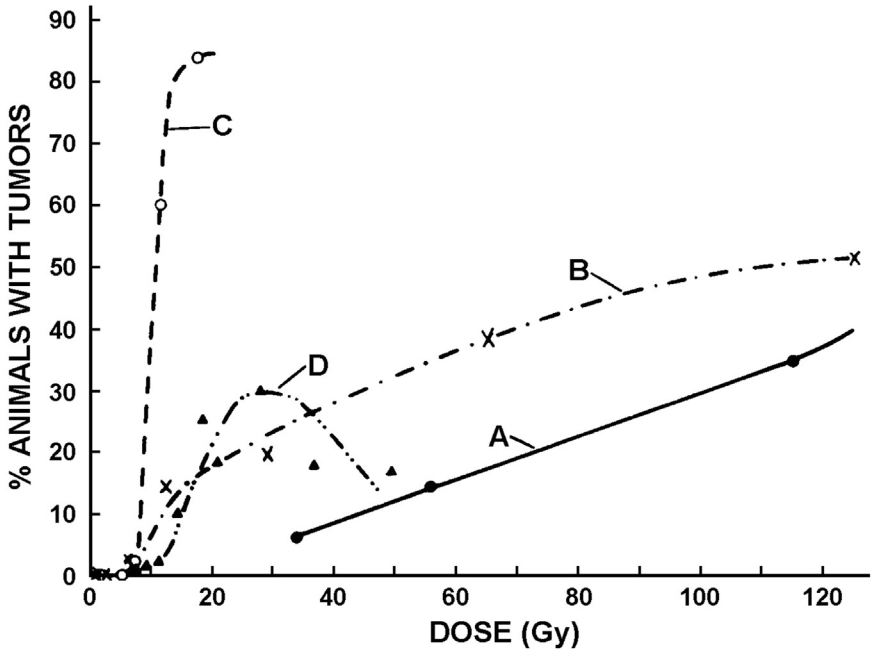


Figure 15.8 Tumor frequency as a function of absorbed dose; see text.

1968). Figure 15.8 seems to indicate a threshold dose at $\leq$5 Gy below which no effect is observed. However, some animal experiments show what looks like a small beneficial effect at small radiation doses, see Figure 15.12.

15.6.4. Mental retardation

In Hiroshima and Nagasaki, both severe mental retardation and lower intelligence levels (obtained from intelligence tests) occurred following prenatal exposure during the 8–15 weeks after pregnancy. In the following 16–25 weeks after ovulation, lesser vulnerability was observed. The exposure to 1 Gy during the early period increased the frequency of mental retardation 40% (normal frequency about 0.8%), and lowered IQ by 25–30 points. No cases of severe mental retardation has been observed at exposures $<$0.5 Gy. No radiation effects on the brain have been observed at low dose rates.

15.7. RADIATION STERILIZATION

Figure 15.2 and Table 15.5 illustrate that at appropriate dosages radiation is an effective means of destroying higher organisms with less harm to simpler ones. This has a practical consequence because radiation can be used for the conservation of food in a quite different manner than the conventional methods of heat, canning, and freezing.